CSCI E-102 Econometrics and Causal Inference with R
Part I of Assignment 8

Let Y denote a binary variable that takes values 0 and 1. Assume we have $n = 900$ observations of X and Y and run the Logistic regression

$$\mathrm{E}[Y|X] = g^{-1}(\beta_0 + \beta_1 X), \quad \text{where } g^{-1}(z) \doteq \frac{1}{1+e^{-z}},$$

that results in the following Maximum Likelihood Estimators (MLEs):

$$\hat{\beta}_0 = -1.2 \quad \text{and } \hat{\beta}_1 = 0.7.$$

Recall that

$$\sqrt{nI(\beta_1)}(\hat{\beta}_1 - \beta_1) \xrightarrow{D} N(0,1) \quad \text{as } n \to \infty,$$

where β_1 denotes the true value of the parameter and $I(\beta_1)$ is the Fisher information.

Assuming $n = 900$ is large enough for the normal approximation, please answer:

1. Test $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$ at a confidence level of $\alpha = 0.05$, if Fisher information $I(0)$ is estimated to be 0.01. Provide the test statistic and p-value. What is your conclusion?
   Hint: Under H_0, the true value of parameter β_1 is 0.
2. Find 95% confidence interval for β_1, if Fisher information $I(\hat{\beta}_1)$ is estimated to be 0.04.
   Hint: $I(\beta_1)$, where β_1 is the true value of the parameter, can be estimated via $I(\hat{\beta}_1)$.

SOLUTION:

1) Given:

$\beta_1 = 0$

$I(\beta_1) = 0.01$

$n = 900$

$\hat{\beta}_1 = 0.7$

$\hat{\beta}_0 = -1.2$

$$TS = \sqrt{nI(\beta_1)} \cdot (\hat{\beta}_1 - \beta_1)$$

$$TS = \sqrt{900 \cdot 0.01} \cdot (0.7 - 0)$$

$$TS = \sqrt{9} \cdot (0.7)$$

$$TS = 3 \cdot 0.7$$

$$TS = 2.1$$

$$\text{p-value} = 2 \cdot P(z > 2.1)$$

$$\text{p-value} = 2 \cdot 0.0179 = 0.0358$$

Since the p-value is less th 0.05, we can reject H_0 and say that $\beta_1 \neq 0$.

2) Given:

$I(\hat{\beta}_1) = 0.04$

$z_{\alpha/2}$ where $\alpha = 0.05 \Rightarrow 1.96$

$$CI = \hat{\beta}_1 \pm z_{\alpha/2} \cdot SE(\hat{\beta}_1) \longrightarrow SE(\hat{\beta}_1) = \frac{1}{\sqrt{nI(\hat{\beta}_1)}}$$

$$CI = 0.7 \pm 1.96 \cdot \frac{1}{\sqrt{900 \cdot 0.04}}$$

$$CI = 0.7 \pm 1.96 \cdot \frac{1}{\sqrt{36}}$$

$$CI = 0.7 \pm 1.96 \cdot \frac{1}{6}$$

$CI_{lower} = 0.7 - 0.32\overline{6} = 0.37\overline{3}$

$CI_{upper} = 0.7 + 0.32\overline{6} = 1.02\overline{6}$